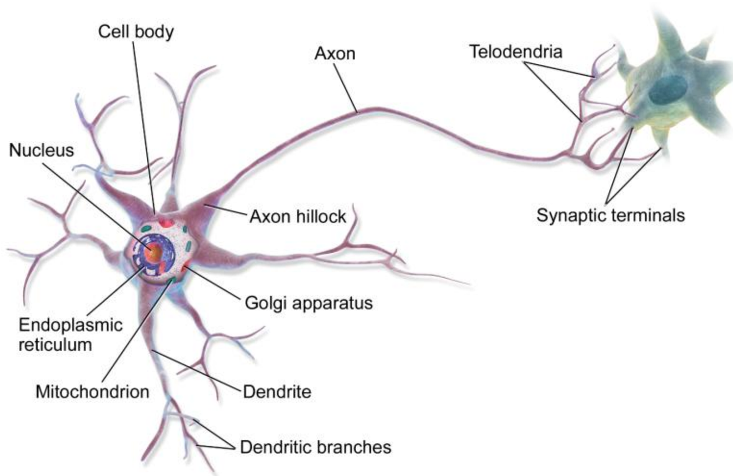
Deep Learning

Introduction to Deep Learning



(Source: Wikipedia)

Chapter One

Basic Introduction to Deep Learning

Shaon Khan
BSc in IIT, JU

Contents

Contents

Chapter One: Basic Introduction to Deep Learning

1.1 When Machines Start Learning on Their Own

We never teach a child to recognize a cat by giving a mathematical definition — "four legs, one tail, ears of a certain shape, so it must be a cat." Instead, we show the child cats again and again, and within a short time, the child learns, all on its own, to recognize cats of many different colors, sizes, and postures. In the history of computer science, we long walked the opposite path — every rule had to be written out separately and explained to the machine. Deep learning has broken this old habit.

The question is: how can a machine learn from experience, the way the human brain does? In this chapter, we will look for the answer to this exact question — starting from the most basic ideas and moving up to the mathematical foundation.

1.2 What is Deep Learning?

In this age of modern technology, artificial intelligence (AI) is rapidly transforming human life. Machine learning (ML) is an important branch of this vast field, and deep learning (DL) is a revolutionary method born out of machine learning itself.

> **Definition**
>
> Deep learning is a branch of machine learning in which a machine uses a multi-layered artificial neural network to learn, on its own, to recognize complex patterns and make decisions from huge amounts of data — without any fixed rules (if-else rules).

This is a completely different approach from traditional programming. In traditional programming, the engineer must write a specific rule for every decision; in deep learning, the algorithm is instead given a large amount of data, and it learns, on its own, to analyze that data and make decisions.

Everyday example: in a traditional algorithm, recognizing a cat would require writing precise mathematical descriptions of ear shape, eye position, whisker pattern, and so on. In the deep learning approach, if you simply feed thousands of cat images as input, the algorithm learns the basic features of a cat on its own — just as a child learns.

1.3 Why Wasn't Traditional Machine Learning Enough?

Traditional machine learning algorithms such as decision trees, support vector machines, and logistic regression once brought about a revolution in data analysis. But when the type of data changed from simple spreadsheets to complex, unstructured forms such as images, audio, or human language, these methods faced one limitation after another.

1.3.1 The Problem of Manual Feature Engineering

Traditional machine learning cannot directly analyze raw data (raw pixels, raw audio waves). Before feeding data into the model, experienced engineers had to manually extract the necessary features — for example, writing code to describe ear shape or texture in order to recognize a cat's picture. If any important feature was missed, the whole model would fail.

Deep learning's multi-layered network solves this problem through fully automatic, end-to-end feature extraction. The first layers learn simple edges, the middle layers combine these into shapes, and the final layers recognize the complete object from those shapes — without any human intervention.

1.3.2 The Challenge of Unstructured Data and High Dimensionality

Traditional ML works wonderfully on well-structured, tabular data (CSV files, spreadsheets, SQL). But even a single HD image contains millions of pixels. If such a huge input is processed as one-dimensional data, the model loses the spatial structure of the image and its performance drops. Architectures such as convolutional neural networks (CNN), recurrent neural networks (RNN), and transformers were built specifically for this kind of unstructured data.

1.3.3 Scale and Complex Nonlinear Relationships

The capacity of traditional models plateaus after a certain point — no matter how much data is added, accuracy stops improving. Deep learning models, on the other hand, keep improving as they get more data and more layers. In addition, by adding nonlinear activation functions such as ReLU, they become able to approximate any complex nonlinear relationship (universal approximation).

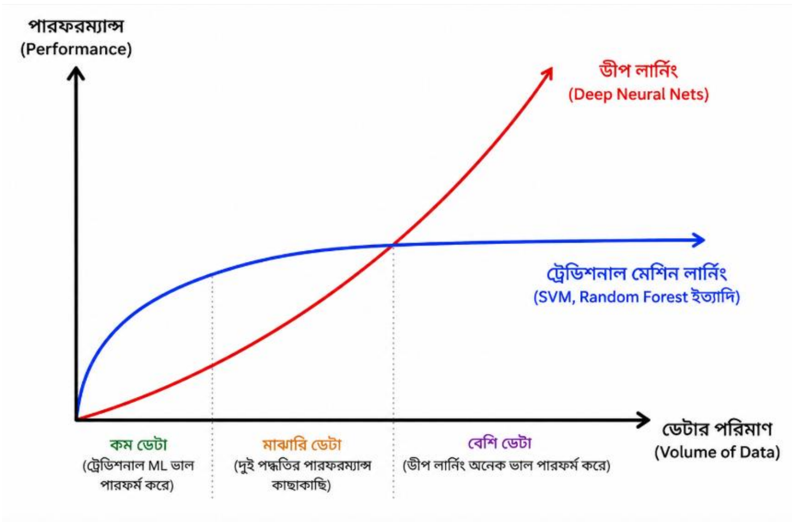


Figure 1.2: As the amount of data increases, the performance of deep learning models keeps improving, while traditional machine learning plateaus after a certain point.

1.3.4 Comparative Summary: Machine Learning vs Deep Learning

Criterion	**Traditional Machine Learning (ML)**	**Deep Learning (DL)**
Best use case	Structured / tabular data	Unstructured data (images, audio, text)
Feature extraction	Manual, depends on human experience	Fully automatic
Data requirement	Works well even on small-to-medium datasets	Requires huge datasets
Hardware	Runs on ordinary CPUs	Needs GPU/TPU for parallel computation
Interpretability	High (transparent logic)	Low (“black box”)

Criterion	Traditional Machine Learning (ML)	Deep Learning (DL)
Training time	Seconds to hours	Hours to weeks

So, is ML now irrelevant? No. Where data is well-structured, computing budget is limited, or a clear explanation of the decision is required (such as credit scoring), ML is still the sensible choice today. DL is needed when data is complex, unstructured, and manual feature creation is impossible or very difficult — such as image recognition, language translation, or real-time bio-signal analysis.

1.4 Inspiration from the Brain: The Story of Biological Neurons

The idea of deep learning is not a new invention — its roots lie in nature itself, specifically in the structure of the human brain. The human brain is made up of billions of tiny cells called neurons, which exchange information with one another through electrical and chemical signals. When a child sees an object for the first time, a new connection pathway forms in the brain; seeing the same object again and again strengthens that connection, and gradually the child learns to recognize the object.

A biological neuron has three main parts — dendrites, which receive signals from nearby neurons; the soma, which gathers all the signals together; and the axon, which sends the final signal on to the next neuron. An important principle operates here — the All-or-None Principle. If the sum of the received signals crosses a certain threshold, the neuron fires; otherwise, it stays completely inactive.

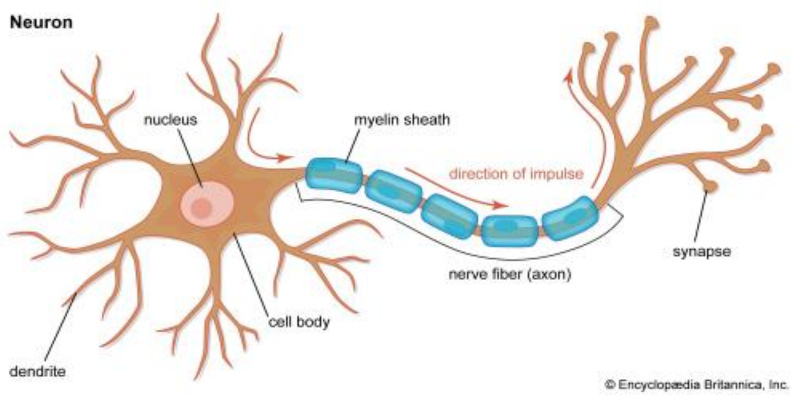


Figure 1.3: Structure of a biological neuron — dendrites, soma, and axon (source: Britannica)

In 1943, scientists Warren McCulloch and Walter Pitts showed that a biological neuron can behave like a logic gate — when several inputs are active together and cross the threshold, it works like an AND gate, and when it fires as soon as any single input is strong enough, it behaves like an OR gate. This discovery laid the first theoretical foundation of artificial neural networks.

Turning this biological process into a mathematical model gave rise to the artificial neural network (ANN) — the cornerstone of today's deep learning.

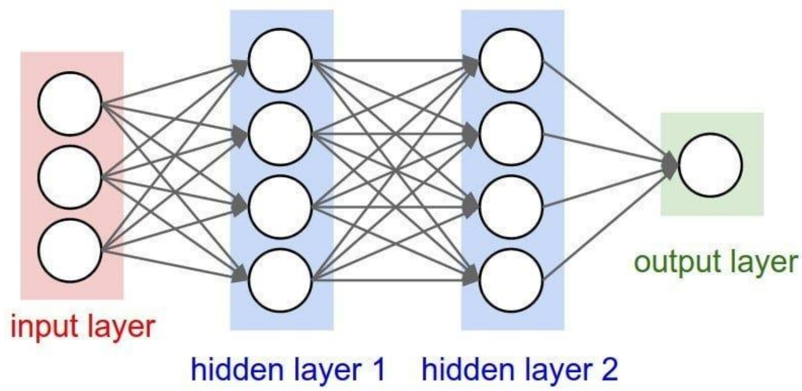


Figure 1.4: Structure of a simple artificial neural network — input, hidden, and output layers.

1.5 How Does Deep Learning Work?

1.5.1 Layered Architecture

The word "deep" comes from the multi-layered structure inside the network. A network usually has three types of layers — the input layer, where raw data enters; the hidden layers, where the actual learning happens (the more hidden layers a network has, the "deeper" it is); and the output layer, which gives the final prediction.

1.5.2 Automatic Feature Extraction

The first hidden layer learns simple features, such as straight lines or curves in an image. The following layers combine these lines to form more complex shapes (such as a round eye). The final layers combine these shapes to recognize the complete object (such as a cat's face). The deeper the data goes, the more the model can move from simple details to complex ideas.

1.5.3 The Learning Process: Predict, Compare, and Correct

Deep learning mainly learns in three steps.

1. Predict: At first, the network gives an output based on a random guess, which is almost always wrong.
2. Compare: The prediction is compared with the actual answer to work out the amount of error.
3. Correct and Learn: The error signal is sent backward through the network, from the end toward the beginning, and the strength of the connections between neurons (the weights) is adjusted.

When this cycle is repeated thousands of times over millions of data points, the amount of error gradually approaches zero, and the model becomes able to make decisions very accurately. Deep learning is, therefore, not some kind of magic — it is the mathematical form of a simple idea: gaining experience, learning from mistakes, and constantly improving.

1.6 Perceptron: The Mathematical Form of the First Artificial Neuron

In 1957, Frank Rosenblatt turned the idea of the biological neuron into a mathematical form and invented the perceptron — the most basic unit of an artificial neural network. An artificial neuron has three parts that correspond to the biological ones:

- Inputs ($x_1, x_2, \ldots, x_n$): the data or features that are received.

- Weights ($w_1, w_2, \ldots, w_n$): show the importance of each input — equivalent to the connection strength of a biological synapse.
- Bias (b): the neuron's internal tendency, which determines how easily the neuron fires ($b = -\text{Threshold}$).

The neuron first calculates the weighted sum of the inputs and weights:

$$z = \sum_{i=1}^{n} x_i w_i + b$$

Here x_i is the i-th input, w_i is its corresponding weight, and b is the bias. Once the sum z is found, it is passed through an activation function, which decides whether the neuron will fire or not. The perceptron uses the step function, which follows the biological all-or-none principle — if z is zero or greater, the output is 1 (the neuron has fired); otherwise, the output is 0.

1.6.1 Weight and Bias Update Process (Perceptron Learning Rule)

How a perceptron learns from its mistakes is determined by the Perceptron Learning Rule, which later laid the foundation for backpropagation and gradient descent. If the model's prediction is ŷ and the actual target is y, the error is calculated as:

$Error = y - \hat{y}$

Since y and ŷ can each only be 0 or 1, the error can take three possible values: Error = 0 (correct guess, no change needed), Error = 1 (the input needs to be strengthened), and Error = −1 (the input needs to be weakened). Each input connection's weight is then changed according to the formula below:

Weight Update Formula

$$w_i^{(\text{new})} = w_i^{(\text{old})} + \Delta w_i$$

Change in Weight

$$\Delta w_i = \eta\,(y - \hat{y})\,x_i$$

Written together

$$w_i^{(\text{new})} = w_i^{(\text{old})} + \eta\,(y - \hat{y})\,x_i$$

Here η (eta / learning rate) is the speed of learning, or the size of the step (usually between 0.01 and 0.1); $(y - \hat{y})$ shows the direction and size of the error; and x_i is the value of the corresponding input — if the input is zero, that weight does not change at all. Since the bias's input is always taken as 1, the input has no effect on the bias update:

$$b^{(\text{new})} = b^{(\text{old})} + \eta\,(y - \hat{y})$$

1.6.2 A Practical Example: Teaching an AND Gate

Suppose we want to teach a perceptron the AND gate, whose rule is — the output is 1 only when both inputs are 1; in every other case the output is 0. Assume, to begin, $w_1 = 0.3$, $w_2 = 0.4$, bias $b = -0.5$, and learning rate $\eta = 0.1$.

Step 1 — Input (1, 1), actual answer y = 1

Calculating the weighted sum (linear combination), we get:

$z = (1 \times 0.3 + 1 \times 0.4) - 0.5 = 0.2$

Since $z = 0.2 \geq 0$, according to the step activation function, the neuron's output will be

$\hat{y} = 1$

Now, finding the error between the actual target (true label) and the prediction, we get:

$Error = y - \hat{y} = 1 - 1 = 0$

That is, the model has predicted correctly. Since the error is zero, there is no need to update the weights or bias. The current values remain unchanged for the next step.

Step 2 — Input (1, 0), actual answer y = 0 (assume, for now, $w_1 = 0.6$, $w_2 = 0.4$, $b = -0.5$)

Calculating the weighted sum (linear combination), we get:

$z = (1 \times 0.6 + 0 \times 0.4) - 0.5 = 0.1$

Since $z = 0.1 \geq 0$, according to the step activation function, the prediction will be

$\hat{y} = 1$

But the actual target (true label) is

$y = 0$

So, calculating the error, we get:

$Error = y - \hat{y} = 0 - 1 = -1$

A negative error shows that the model has wrongly predicted the positive class. So, according to the Perceptron Learning Rule, the weights and bias must be updated.

Updating weight w_1:

$\Delta w_1 = \eta\,(y - \hat{y})\,x_1 = 0.1 \times (-1) \times 1 = -0.1$

$w_1(new) = 0.6 + (-0.1) = 0.5$

Updating weight w_2:

Since $x_2 = 0$,

$\Delta w_2 = 0.1 \times (-1) \times 0 = 0$

So,

$w_2(new) = 0.4$

Updating the bias:

$\Delta b = 0.1 \times (-1) = -0.1$

b(new) = −0.5 + (−0.1) = −0.6

So, after the update, the new parameters are:

$w_1 = 0.5$, $w_2 = 0.4$, $b = -0.6$

That is, because of the wrong prediction, the Perceptron Learning Rule has adjusted the weights and bias so that, in the next iteration, the model has a better chance of making the correct decision.

By running this cycle again and again (epochs) over the whole dataset, the perceptron reaches a set of weights and bias where Error = 0 for every input — meaning the model has fully learned the logic of the AND gate.

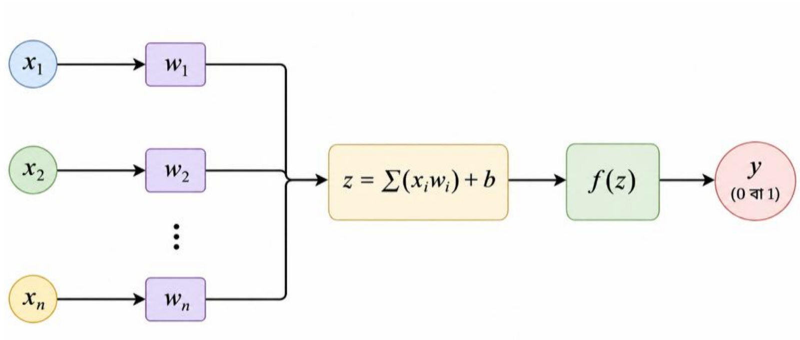


Figure 1.5: The mathematical structure of a perceptron — the combination of inputs, weights, bias, and the activation function.

1.7 Real-Life Applications of Deep Learning

Deep learning is no longer just a subject for research labs. Because of its ability to analyze complex, unstructured data (images, audio, video, text) and find patterns in it, it is today the silent driving force behind many technologies in our everyday lives.

Healthcare

Convolutional neural networks (CNN) are used to detect tumors or bone fractures by analyzing X-rays, MRIs, and CT scans. Beyond this, noisy bio-signals such as ECG or EEG can be cleaned up to predict heart disease, and models such as AlphaFold can estimate protein structures, which is helping to speed up drug discovery.

Business & E-Commerce

The recommendation engines on YouTube, Netflix, or Amazon that suggest new content based on our preferences are built on deep learning. In finance, it analyzes transaction patterns to detect fraud within moments and supports algorithm-based trading decisions.

Natural Language Processing (NLP)

Models based on the transformer architecture — such as modern chatbots and virtual assistants — can understand the context of human language to give answers, translate instantly, and convert speech into written text.

Image and Computer Vision

Face unlock on smartphones, biometric checks at airports, and detecting suspicious activity in CCTV footage — all of these are real-world applications of computer vision, where a decision is made by analyzing every pixel of an image.

Autonomous Vehicles

In self-driving cars, deep learning is used to analyze information from cameras, lidar, and radar in order to detect pedestrians and other vehicles, and it also makes predictive maintenance possible by analyzing sensor data from vehicle parts.

Field	Main Technology / Model	Main Purpose
Healthcare	CNN, Autoencoder	Disease detection and signal cleaning
Autonomous vehicles	CNN, Transformer	Real-time object detection
Finance	Anomaly detection network	Fraud prevention and market analysis
Language & voice	Transformer (Attention)	Voice assistants and translation
E-commerce	Deep collaborative filtering	Personalized content suggestions

1.7.1 Hardware Context: CPU vs GPU

Because deep learning models need to perform a huge number of matrix calculations in parallel, specialized hardware is required.

Feature	CPU	GPU
Structure	Fewer but more powerful cores	Thousands of small cores
Suitability	Sequential calculation	Parallel calculation
Use in deep learning	Small models, pre/post-processing	Training and inference of large models

Chapter Summary

- Deep learning is a branch of machine learning in which a machine uses multi-layered neural networks to learn patterns on its own from huge amounts of data.
- The neurons in the human brain, and their threshold-based firing process, are the main inspiration behind artificial neural networks.
- Traditional machine learning is limited on unstructured data because of manual feature engineering and limited scalability; deep learning overcomes this limitation through automatic feature extraction.
- The perceptron is the most basic unit of an artificial neural network, working through the equation $z = \Sigma x_i w_i + b$ and an activation function.

- Through three steps — predict, compare, and correct — a network gradually reduces error and learns to make correct decisions.
- Deep learning is used extensively in almost every area of real life — healthcare, business, language processing, computer vision, and autonomous vehicles.

Key Definitions

- Deep Learning: A branch of machine learning based on multi-layered neural networks that learns features automatically.
- Neuron: The basic unit that receives, analyzes, and transmits information — in both the biological and artificial sense.
- Perceptron: A single-layer artificial neuron made up of inputs, weights, bias, and an activation function.
- Activation Function: The function that determines whether a neuron will fire or not.
- Learning Rate (η): The parameter that controls how much the weights change at each step.

Key Formulas

- **Weighted Sum:** $z = \sum_{i=1}^{n} x_i \, w_i + b$
- **Error Calculation:** Error $= y - \hat{y}$
- **Weight Update Rule:** $w_i^{(\text{new})} = w_i^{(\text{old})} + \eta \, (y - \hat{y}) \, x_i$
- **Bias Update Rule:** $b^{(\text{new})} = b^{(\text{old})} + \eta \, (y - \hat{y})$

Common Interview Questions

1. What is the main difference between machine learning and deep learning?
2. Why can't a perceptron solve the XOR problem?
3. What effect does a very large or very small learning rate have on training?
4. Why is a deep learning model often called a "black box"?
5. Why is a GPU more effective than a CPU for training deep learning models?

Multiple Choice Questions (MCQ)

1. What kind of structure does deep learning mainly rely on?

   (a) Decision tree
   (b) Multi-layered artificial neural network
   (c) Linear regression
   (d) K-Means clustering

2. What is the bias (b) of a perceptron mathematically equivalent to?

   (a) Learning rate
   (b) Negative threshold
   (c) Number of inputs
   (d) Activation function

3. Which of the following is NOT an example of unstructured data?

   (a) Image

(b) Audio recording

(c) Excel spreadsheet

(d) Natural language text

Short Questions

4. What does the "All-or-None Principle" mean?

5. How does automatic feature extraction set deep learning apart from traditional ML?

6. What is the role of the learning rate (η)?

Review Questions

1. Using a real example, explain why deep learning is more effective than traditional machine learning for image recognition.

2. Explain the mathematical structure of a perceptron, and show how it learns from mistakes by updating its weights and bias.

3. Compare the applications of deep learning in healthcare and finance.

Practical Exercise

Assume initial weights $w_1 = 0.2$, $w_2 = 0.2$, bias $b = -0.2$, and learning rate $\eta = 0.1$. To teach a perceptron the OR gate, work out, step by step, the weight and bias updates for the first two inputs (0,1) and (1,0), and write down the values of z, $\hat{y}$, and Error after each step.